Radio interferometry imaging with Difmap

Miguel Pérez-Torres

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CSIC
Consejo Superior de Investigaciones Científicas

IAA
Instituto de Astrofísica de Andalucía

Excelencia Severo Ochoa

Difmap Cheat Sheet

Starting up

- `difmap` - (from terminal)
- > `help` - use it. It is concise and to the point.
- > `difmap.log_*` - (to execute as command line from within Difmap
- > `exit` - it does what you think

Reading, examining and editing the UV data

- > `obs FILENAME[.uvf]` - read in the data
- > `select I` - we'll use total intensity (Stokes I)
- > `radplot` - Plot amplitude vs u-v radius. Select "/xw", when prompted
- > `uvplot` - Plot u-v coverage
- > `projplot` - Look at a cut of amplitude and/or phase along a radial line in the UV plane.
- > `vplot 9` - Plot 9 baselines per page. That's convenient for VLBA obs-ns, but maybe too many to be see well in a laptop, so experiment with other numbers.
- > `wobs FILENAME.edt` - If you edit your data, you need to save it.

Difference mapping - Setting it up

- > `startmod` - For most VLBI sources, a point source model is a good start
- > `mapsize 1024, 0.1` - Default units are milli-arcseconds in the image plane, and Megalambda in the uv-plane. The number of pixels in the exaple is 1024, each one of 0.1 milliarseconds. Remember to use at least three pixels across the synthesized beam.
- > `uvweight 2, -1` - uniform weighting (default; provides higher resolution, but worse sensitivity)
- > `uvweight 0, -1` - natural weighting (provides poorer resolution, but better sensitivity).

Difference mapping - Cleaning (finally!)

- > `mapplot` - First time you run it, it will give you a 'dirty' map. Enter the window and draw a first window where Difmap will clean.
- > `clean 200, 0.05` - Start with a relatively small gain (¡=0.1), and not many iterations: clean 100, 0.1; clean 200, 0.05 are usually good starts.
- > `mappl` - Look at your residual data and judge if you should continue cleaning in that window, or adjust by removing/adding new windows.
- > `selfcal` - When you've cleaned enough, selfcalibrate the phases of your visibility data.
- > `mappl` - Look again at your residual data and iterate the above scheme until positive and negative residuals are of the same order, approximately. Note that every time the "sigma" value must get lower and lower, your cleaned model should have higher and higher values, and the residuals (both positive and negative) should decrease.
- > `mapplot cln` - Look at the clean map to see how it looks.
- > `radplot, vplot,...` - Look at the visibilities and see whether your model is reproducing the real data.

Final steps

- > `gscale` - Amplitude correction for each antenna. Use only if you have a good model and enough signal-to-noise ratio in the data. This may show that some antenna was poorly calibrated (unlikely for the MOJAVE cases we're discussing in this tuto).
- >`mapplot` - To see the effect of gscale in the data
- >`selfcal true, true, 30` - You can play with those values
- >`uncal false, true` - To undo the amplitude corrections, but not the phase corrections.
- >`save FILENAME-final` - To save your data, models and windows, for future use.

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